

1. OBJECTIVE

The main aim of this EDA is to understand the sales and profit of the Superstore dataset. The analysis looks at categories, sub-categories, discounts, regions, sales, quantity and profit. Different charts were used to find simple patterns in the data.

2. ANALYSIS AND KEY FINDINGS

Analysis	Key Finding	Business Insight
Sales by Category	Sales contribution varies across categories.	High-sales categories can be prioritized for inventory and marketing decisions.
Profit by Category & Sub-Category	Profitability differs across products; some sub-categories may generate low or negative profit.	Loss-making products should be reviewed for pricing and discount policies.
Sales & Profit Distribution	Box plots show variation and extreme values in sales and profit.	Outliers should be investigated to identify unusual transactions or business opportunities.
Discount vs Profit	Profit changes with different discount levels, with higher discounts potentially reducing profitability.	Excessive discounting should be controlled to protect profit margins.
Regional Performance	Profit contribution varies between regions.	Regional pricing, product mix, and discount strategies should be analyzed separately.
Correlation Analysis	Sales, Quantity, Discount, and Profit show different levels of relationships.	Correlation helps identify variables that are useful for further business analysis.

DISCOUNT AND PROFIT ANALYSIS

The **Discount vs Profit** analysis was used to understand the relationship between discount and profit. The results show that profit changes at different discount levels. At higher discount levels, the average profit becomes lower and can also become negative. This shows that giving high discounts can reduce profit.

4. MAJOR BUSINESS INSIGHTS

- Technology had the highest sales at **839,893.2790**.
- Technology also had the highest profit at **146,543.3756**.
- Furniture had **754,747.7613** sales but only **19,729.9956** profit.
- Office Supplies had **731,893.3140** sales and **126,023.4434** profit.

- West had the highest regional profit at **110,798.8170**.
- Average profit became negative from the **30% discount** level.
- The dataset has both negative and positive profit values.

5. RECOMMENDATIONS

- Keep discounts under control, especially at higher discount levels.
- Check products and sub-categories that have low or negative profit.
- Pay attention to Furniture because its sales are high but its total profit is much lower.
- Continue monitoring Technology because it has the highest sales and profit.
- Compare profit between regions regularly.
- Study sales and profit together instead of looking only at sales.

6. CONCLUSION

The EDA helped to understand the sales and profit performance of the Superstore dataset. Technology had the highest sales and profit, while Furniture had much lower profit compared with its sales. The regional analysis showed that West had the highest total profit. The discount analysis also showed that higher discounts can lead to negative average profit. Overall, the analysis gives a clear view of how category, region and discount are related to business performance